

University Applications & Fund Allocation Analysis

Project Overview

This project focuses on analyzing university applications and fund allocation data to identify application trends, evaluate university performance, assess the impact of special grants, and compare funding distribution across institutions. The project includes extensive data transformation using Power Query and dashboard development using Power BI.

Business Objective

The objective of this project is to analyze university application trends, evaluate funding allocation across institutions, identify top and bottom performing universities, and generate actionable insights using interactive Power BI dashboards.

Problem Statement

Problem Statement 1

Perform data transformation and preparation using Power Query by applying various transformation techniques such as append, merge, split columns, pivot/unpivot operations, conditional columns, and data modeling.

Problem Statement 2

Develop interactive Power BI dashboards to analyze university applications, funding allocation, special grants, and demographic distribution of applicants.

Dataset Information

Total Datasets Used: 2

Dataset 1: University Admissions_New

- Records: 4,000 Rows
- Attributes: 23 Columns

Dataset 2: University Applications Charter

- Records: 900 Rows
- Attributes: 22 Columns

The datasets contain information related to university applications, applicant demographics, institutional details, funding allocations, and special grant programs.

Tools & Technologies

- Microsoft Excel
- Power BI
- Power Query
- DAX (Data Analysis Expressions)

Data Transformation Techniques

The following Power Query transformations were applied:

- Append Queries
- Merge Queries
- Split Columns
- Fill Operations
- Conditional Columns
- Index Columns
- Pivot & Unpivot Transformations
- Data Formatting
- Summary Table Creation
- These transformations improved data quality and prepared the datasets for analytical reporting.

Data Modeling & DAX

- Data relationships were established between transformed datasets to support analytical reporting.
- Two custom DAX measures were created to support analytical calculations and dashboard reporting.

Dashboard Development

Two major analytical areas were covered:

Applications Analysis

- Applications Target Trends by State
- Geo-State Applications Dashboard
- Universities & Funds Analysis
- Application Distribution by Race

Universities Analysis

- Top 10 Universities by Applications

- Top 10 Universities with and without Special Grants
- Bottom 10 Universities by Applications
- Applications vs Fund Allocation Analysis

Key Visualizations

- State-wise Application Trend Analysis
- Geographic Application Distribution
- University Funding Analysis
- Race-wise Application Distribution
- Top Performing Universities
- Special Grant Impact Analysis
- Bottom Performing Universities
- Application vs Fund Allocation Comparison

Key Insights

- Significant variation exists in application volumes across universities.
- Certain universities consistently attract higher application volumes than others.
- Special grant programs influence application trends for selected institutions.
- Funding allocation patterns differ considerably among universities.
- Geographic analysis highlights differences in application distribution across states.
- Demographic analysis provides visibility into applicant diversity across institutions.

Outcomes

- Improved understanding of university application trends.
- Enhanced visibility into fund allocation strategies.
- Identification of top and bottom performing universities.
- Better decision-making support through interactive dashboards.
- Streamlined data preparation process using Power Query.
- Developed an end-to-end Business Intelligence solution using Power BI.

Conclusion

This project demonstrates end-to-end Business Intelligence workflow using Power Query, DAX, and Power BI. Through data transformation, modeling, and dashboard development, meaningful insights were generated regarding university applications, funding allocation, institutional performance, and applicant demographics.